



Technische Hochschule
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Fakultät
Maschinenbau

Autonomes Fliegen

II. Various Components of UAVs, Autonomous UAVs and C-UAS

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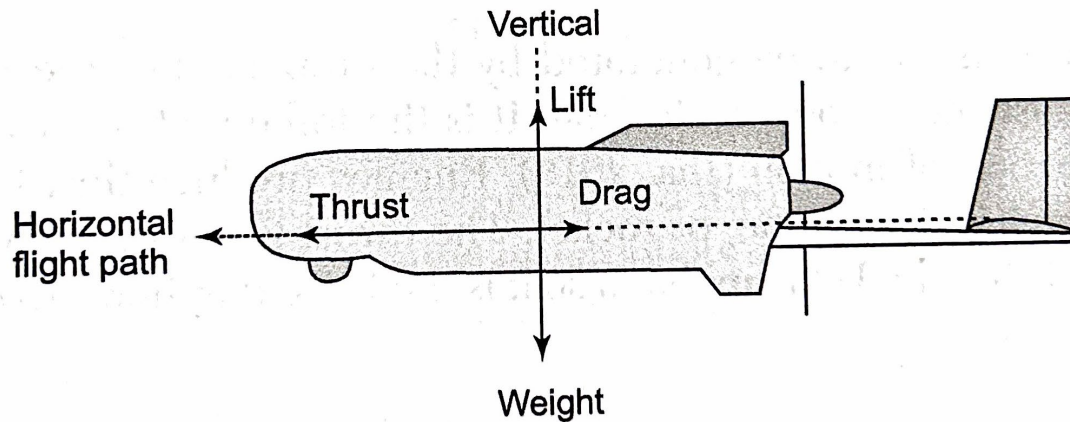
II. Various Components of UAVs and Autonomous UAVs

Content

- **Basic Aerodynamics**
- **Stability and Control of UAV**
- **Components of a UAV**
- **Autonomous UAVs**
- **C-UAS**

Basic Aerodynamics

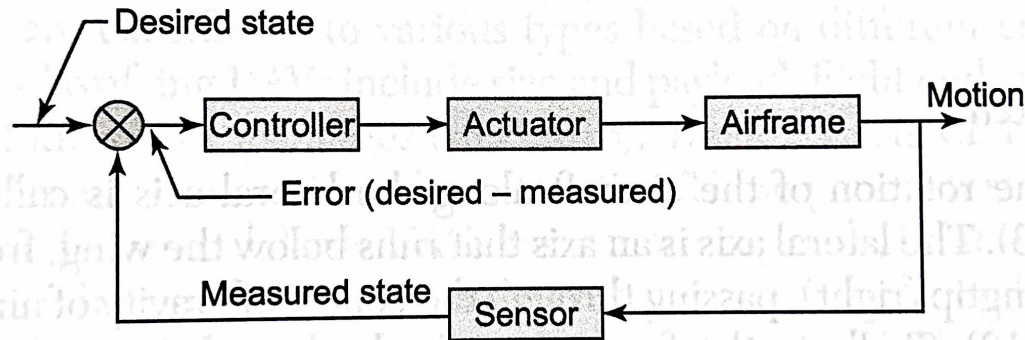
Primary forces acting on an air vehicle



- **Lift:** Is the force that acts at a right angle to the direction of motion through the air => responsible for lifting the UAV up
- **Drag:** Mechanical forces applied by any fluid to oppose the motion of any object called drag
- **Thrust:** Force generated by the propellers of the multirotor to work against the aerodynamic drag

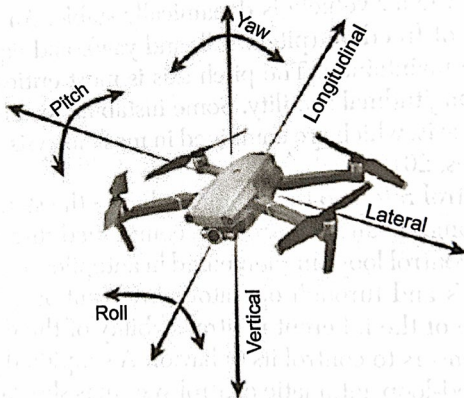
Stability and Control of UAV

Block diagram of a control loop



All UAVs must have

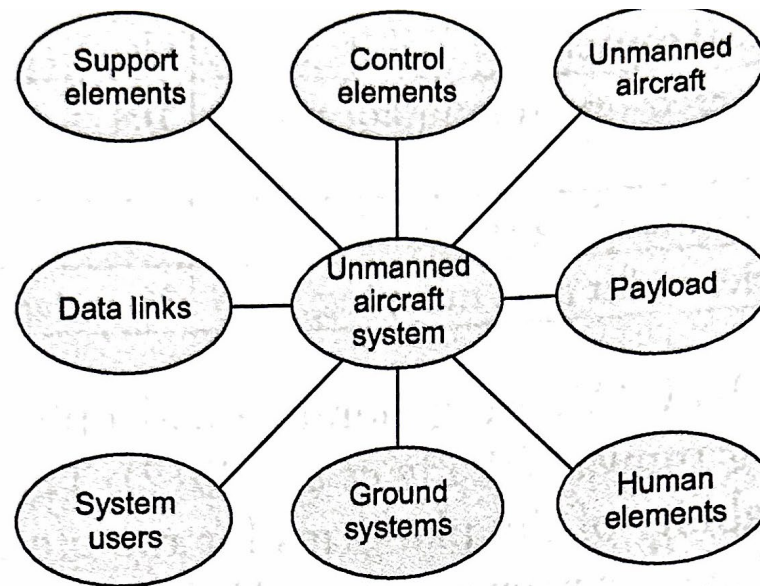
- stability
- controllable



- **Roll:** Rotation of the aircraft along the **longitudinal** axis
- **Pitch:** Rotation of the aircraft along the **lateral** axis
- **Yaw:** The rotation of the air vehicle (left or right) along the **perpendicular** axis

Components of a UAV

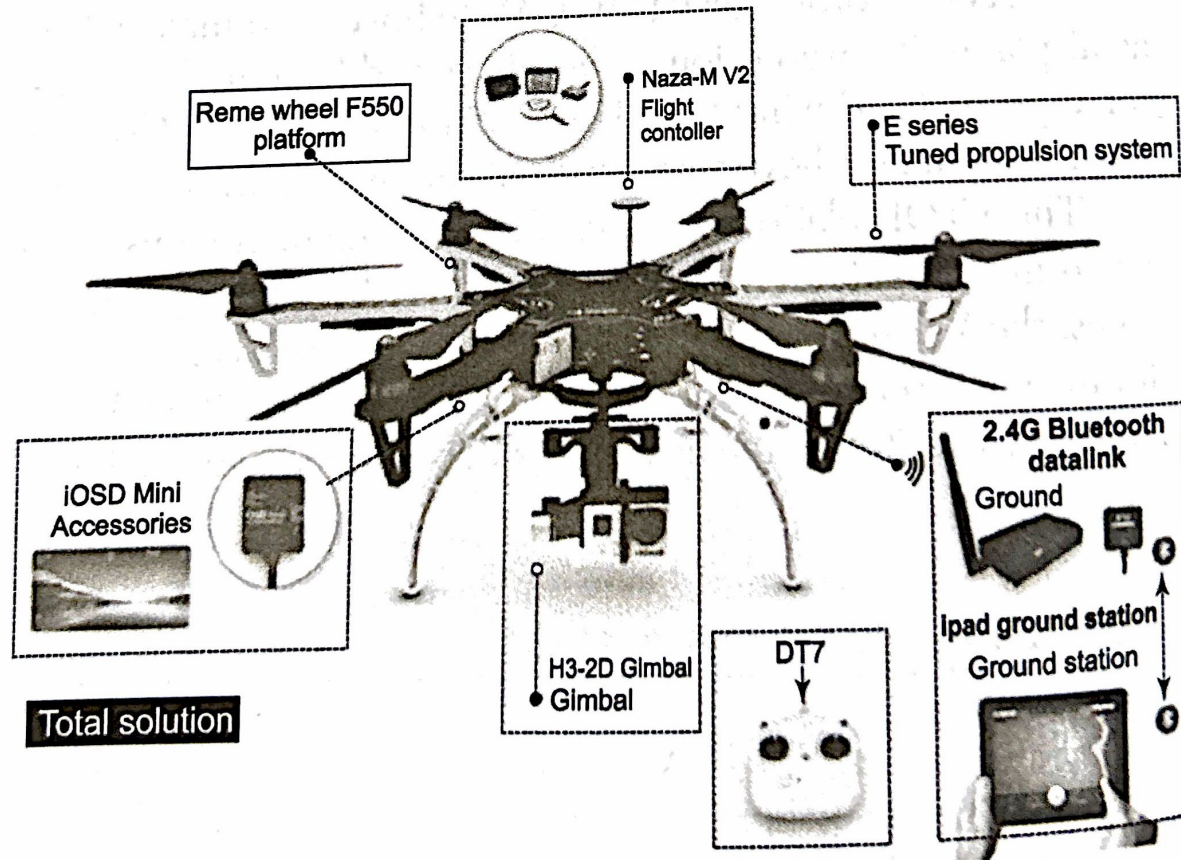
Main Components



Quelle: Book Unmanned Aerial Vehicle. P.K.Garg

Components of a UAV

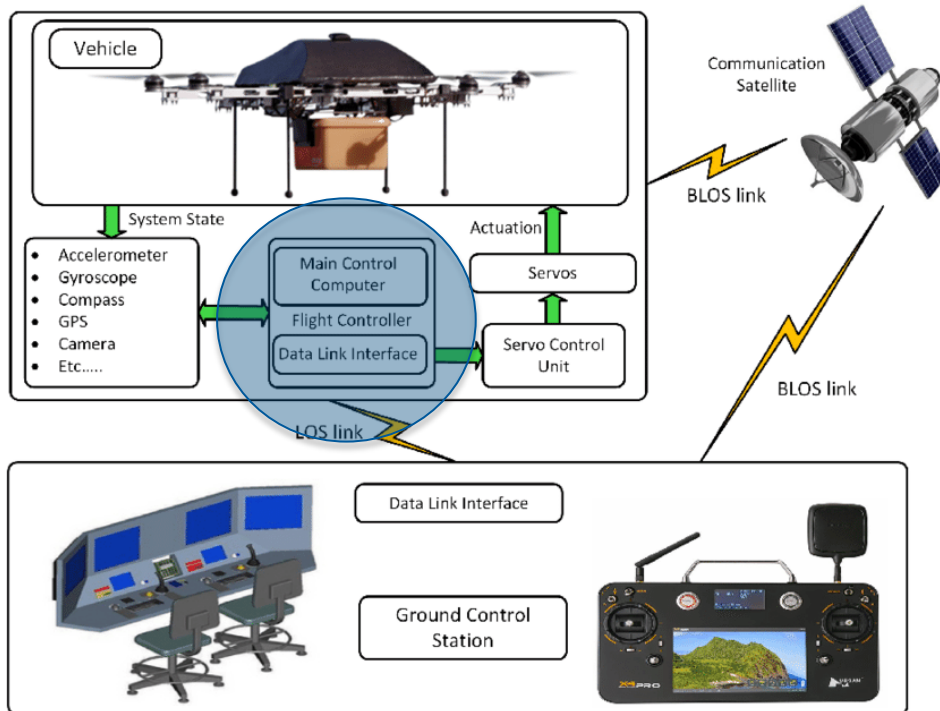
Pictorial view of each component (Uerkwitz et al, 2016)



Quelle: Book Unmanned Aerial Vehicle. P.K.Garg

Components of a UAV

Flight controller and associated links (Altawy et al., 2016)



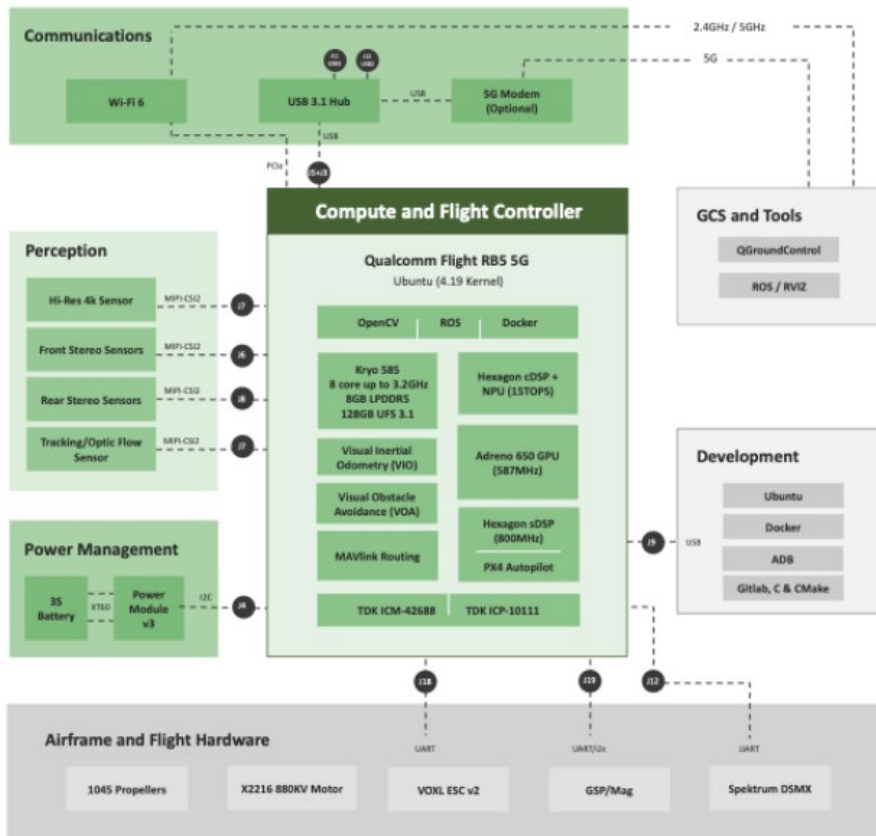
HW of a FCS consists of:

- Main control modul,
- signal conditioning and interface module,
- a data aquisition module
- servo drive module

- **Flight control system (FCS)** is the brain of the aerial vehicle
- Consists of **Hardware** and **Software**
- **FCS** can be programmed to take-off and fly waypoints
- **FCS** collects the flight state data from the sensors in real-time, receives via data link control commands from the GCS.
- For the **stabilization** a Flight state controller generate the outputs to the servos

Components of a UAV

Example: Qualcomm Flight™ RB5 5G Drone Reference Design

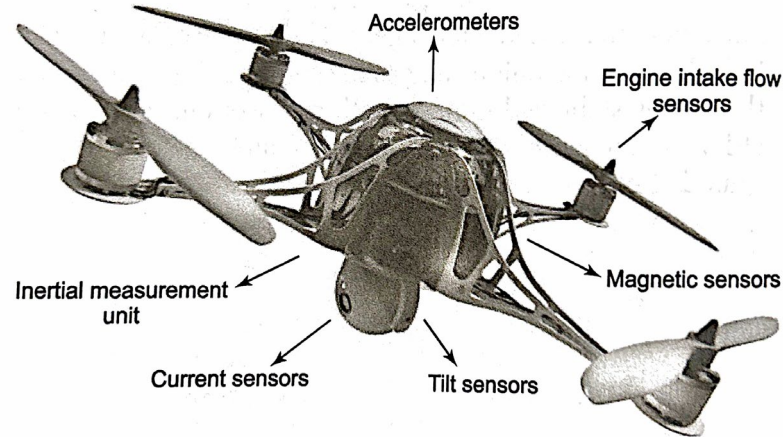


Quelle: Internet

Components of a UAV

Various sensors used in modern UAVs

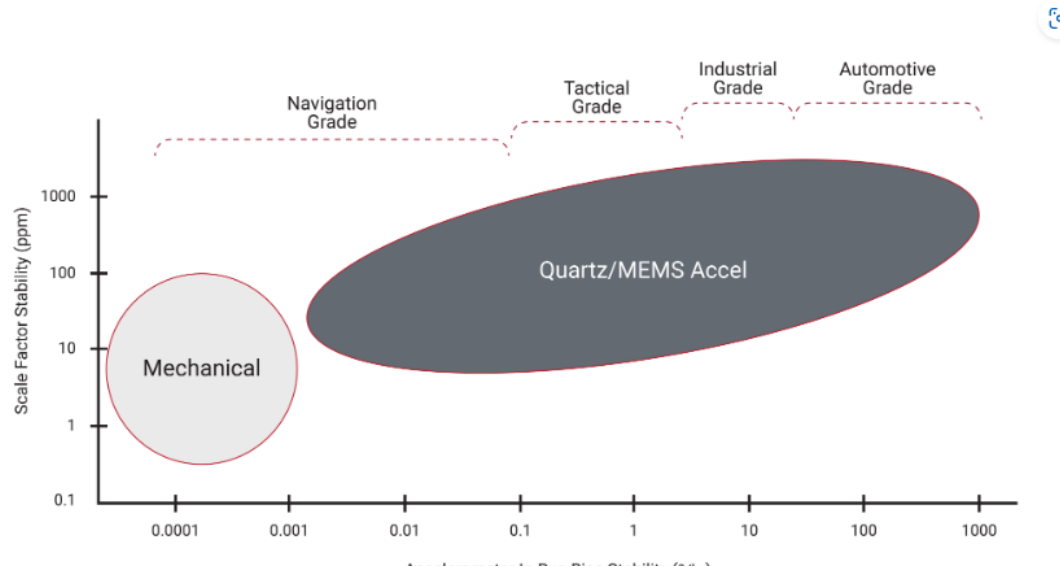
Quelle: Book Unmanned Aerial Vehicle. P.K.Garg



- **Accelerometers**: determine **position** and **orientation** in flight
- **Gyroscope**: part of the Inertial measurement Unit (IMU) also known as angular rates sensor or angular velocity sensors; measures the rate of rotation of an object about its axis (rad/s)
- **Magnetometer**: measures magnetism; Aiding sensor for navigation
- **Tilt sensors**: combined with gyros and accelerometers => maintain the **level of flight**
- **Inertial measurement Unit (IMU)**: Essential component for an UAV; detecting current rate of acceleration; Determines **orientation**, **velocity** and **location**
- **GPS**: Global Positioning System

Components of a UAV

IMU grades



IMU grade	Accelerometer bias		Gyro bias	
	mg	m s^{-2}	$^{\circ} \text{h}^{-1}$	rad s^{-1}
Marine	0.01	10^{-4}	0.001	5×10^{-9}
Aviation	0.03–0.1	3×10^{-4} – 10^{-3}	0.01	5×10^{-8}
Intermediate	0.1–1	10^{-3} – 10^{-2}	0.1	5×10^{-7}
Tactical	1–10	0.01–0.1	1–100	5×10^{-6} – 5×10^{-4}
Consumer	>3	>0.03	>100	> 5×10^{-4}

Quelle: Internet

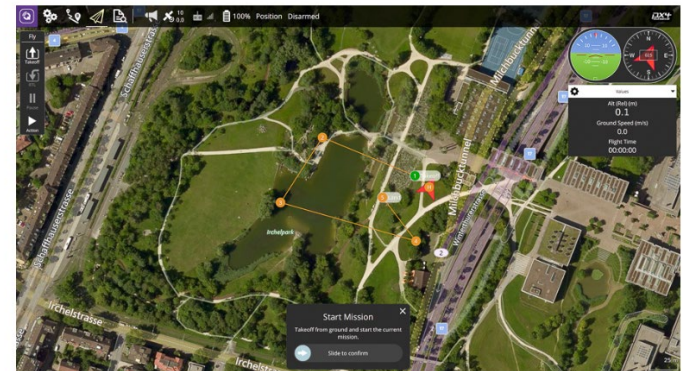
Components of a UAV

Ground Control Station (GCS)

- Also known as **base station (BS)** is used to remotely control the UAV
- **GCS is composed of three parts**
 - Ground monitoring station
 - Remote control transmitter
 - Telemetry receiving antenna
- Vary in size according to type and missions
- Receives telemetry data
- Displays real-time transferred images
- **Computational system designed to**
 - Monitor mission development
 - Eventually Operates the UAV and payload



ground control station for VTOL UAV and Drones building



ground control station for VTOL UAV and Drones building

Quelle: Book Unmanned Aerial Vehicle. P.K.Garg

- **Four types of autonomy related to UAVs (Krishnan (2009))**
 - I. **Tele-operation** – which involves continual use of remote control
 - II. **Pre-programmed autonomy** – which involves a range of pre-programmed computer-directed behaviours
 - III. **Supervised autonomy** – The UAV operators make corrections when any problem occurs with the UAV
 - IV. **Complete Autonomy** – where a robotic/UAV system can identify and handle the problem that arises without any human intervention

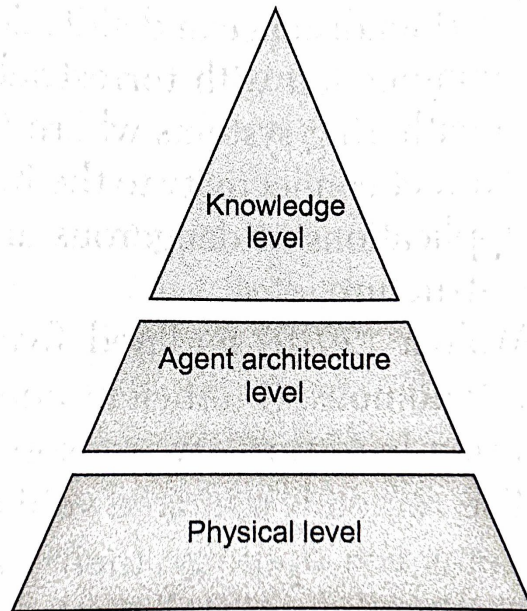
Quelle: Book Unmanned Aerial Vehicle. P.K.Garg

Autonomous UAVs

Architecture of an Autonomous system

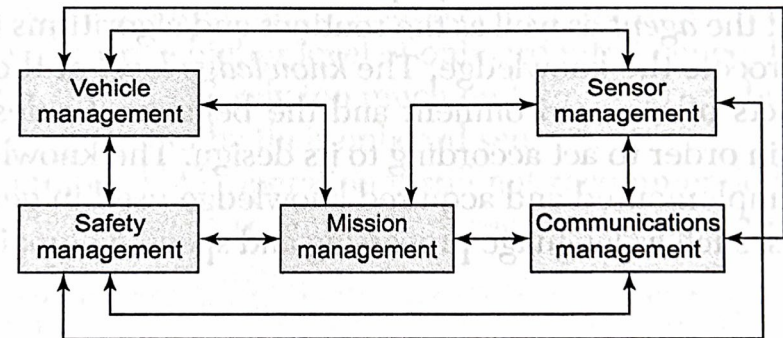


Abstraction hierarchy of an agent (Stenger et al., 2012)



Quelle: Book Unmanned Aerial Vehicle. P.K.Garg

Architecture of an autonomous system

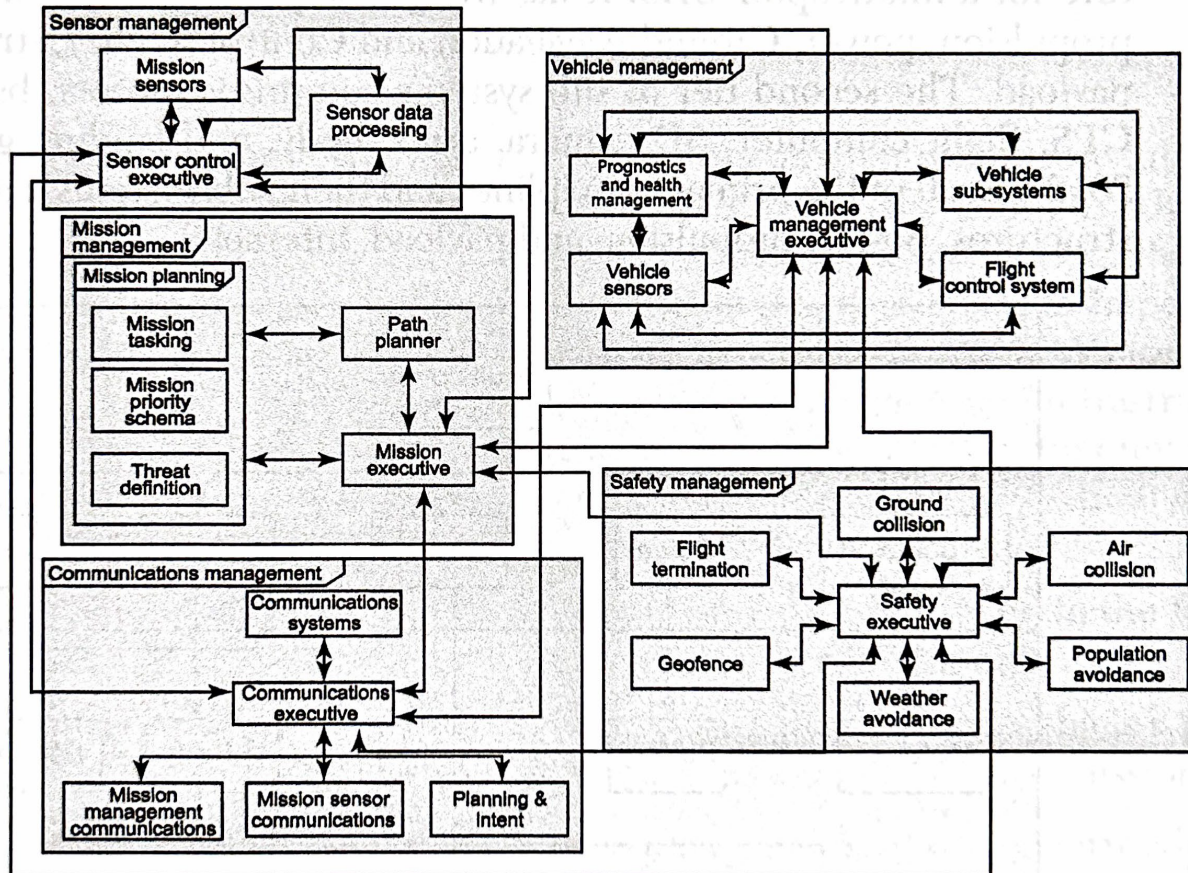


Autonomous UAVs

Architecture of an Autonomous system



■ Detailed system architecture (Eaton et al., 2016)

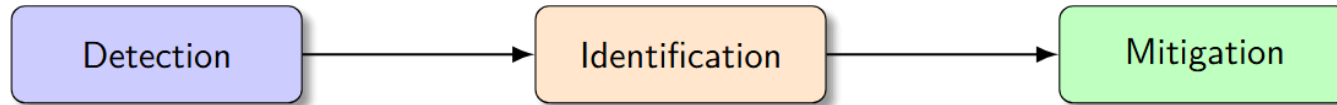


Quelle: Book Unmanned Aerial Vehicle. P.K.Garg

Counter-UAS

Functional Chain, Components, Effectors, Challenges

- Functional chain (detection → identification → mitigation)
- Main components (sensors, fusion, C2, effectors)
- Effectors & engagement logic
- Key challenges & open research
- Application: Critical Infrastructure Protection (CIP)

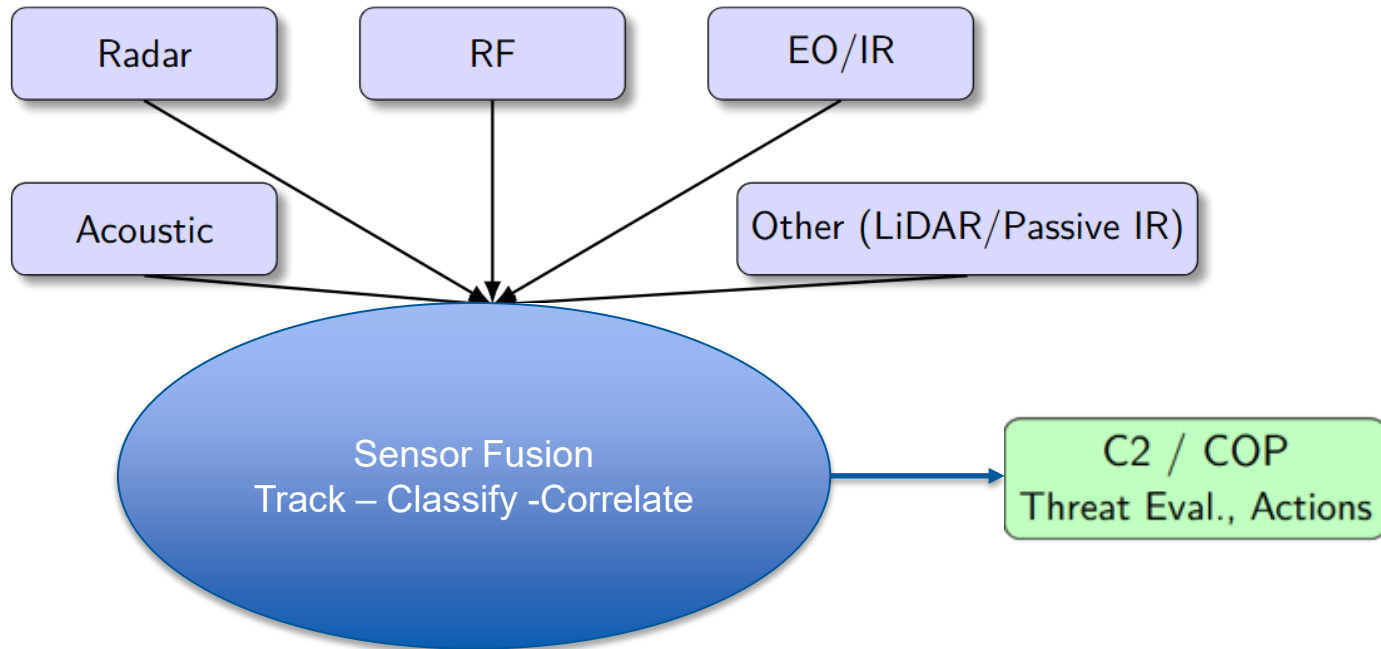


C2 & Sensor Fusion across phases
Automated decision support

More Detailed

- Detect → Identify → Decide (C2) → Mitigate → Assess/Record
- implemented as a layered, multi-sensor architecture: radar, RF, EO/IR, acoustic (and sometimes LiDAR) for surveillance
- fusion & AI for discrimination
- policy-driven C2;
- soft- and hard-kill effectors;
- plus deterrence/legal overlays (e.g., geofencing, Remote ID).

- **Sensors:** Radar, RF, EO/IR, Acoustic, (LiDAR/Passive IR)
- **Fusion:** Track association, classification, confidence
- **C2 (Command & Control):** COP, ROE/policy, tasking, logging
- **Effectors:** Soft-kill (RF/GNSS), DEW (Laser/HPM), Kinetic Interceptors, Cyber

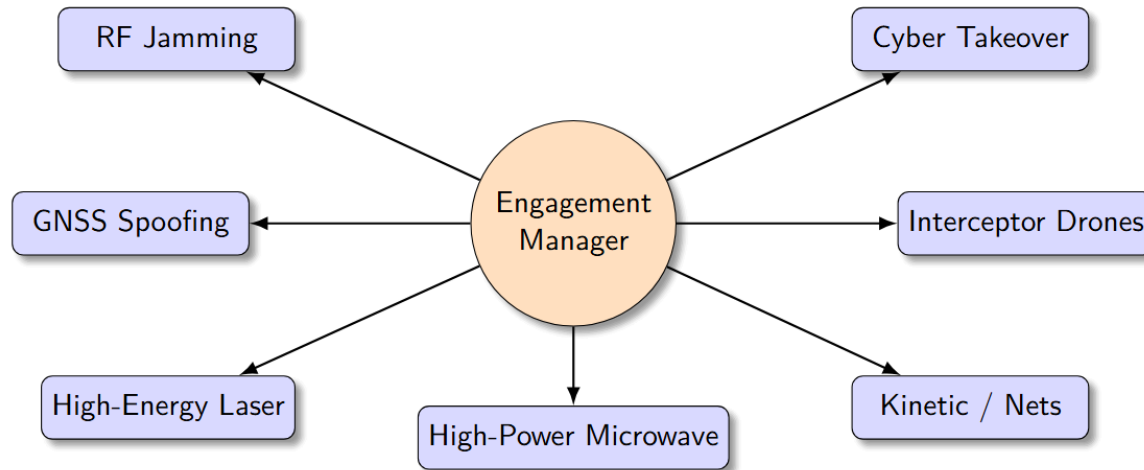


Detection Modalities (Snapshot)



Modality	Strengths	Limitations	Typical Use
Radar	All-weather, long-range	Small RCS & birds, cost	Airports, bases
RF	Passive, pilot location	Autonomous/silent drones	Prisons, venues
EO/IR	Visual confirmation	LoS, weather dependent	Site security
Acoustic	Hidden/indoor cueing	Short range, noise	Urban perimeters

- RF fingerprinting, protocol metadata, Remote ID
- Computer vision (EO/IR), trajectory/behavioral AI
- Confidence scoring via multi-sensor correlation
- Slew-to-cue PTZ, track handover across sensors



- **Electronic countermeasures:** RF link denial, GNSS spoofing/denial; effective against piloted links but weaker against fully autonomous routes.
- **Directed energy:** HEL and HPM for component upset/burn-through; promising against swarms but subject to dwell/atmospherics/safety constraints.
- **Kinetic/interceptor:** Nets, capture drones, projectiles—needed for silent/autonomous targets or RF-dense/legally constrained sites

- **Detect** → Confirm (EO/IR) → Classify (benign/hostile)
- **Decide**: ROE, safety buffer, airspace deconfliction
- **Act**: Soft-kill first; escalate to DEW/kinetic if necessary
- **Coordinate**: ATC/utility control rooms, law enforcement, on-site security
- **Assess**: BDA, forensics, lessons learned

Outer: Detection & Early Warning — Long-range Radar, Passive RF, Wide-area ISR

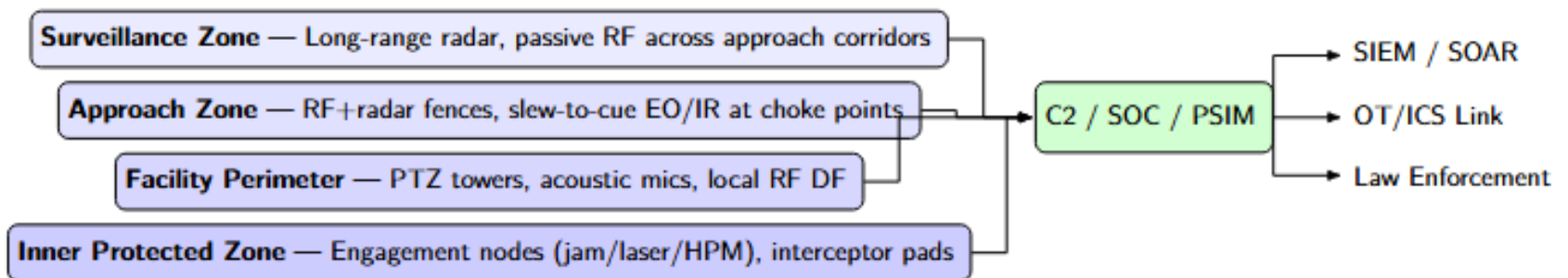
Middle: ID & Tracking — EO/IR PTZ, Tracking Radar, AI Classifier

Inner: Engagement — Jammers, Lasers/HPM, Kinetic, Interceptors

C2 Backbone — Fusion, Policy, ROE, Handover, Logging

- Low-signature LSS drones; radar/EO clutter discrimination
- Autonomous/signal-silent drones; GNSS-denied navigation
- Swarm saturation; target prioritisation and resource management
- Weather/LoS constraints for EO/IR/laser; RF collateral effects
- Integration (open interfaces), cyber-resilience, legal constraints

Protecting Critical Infrastructure (CIP)



■ Detection & tracking

- **Modalities:** RF sensing (controller/UAS emissions & Remote ID), primary/short-range radar (incl. micro-Doppler), EO/IR (PTZ confirmation), acoustic arrays; often combined in practice to cut false alarms.
- **Trends:** Multi-sensor fusion and ML have lifted small-UAS discrimination (e.g., bird vs. quad) and improved cueing to EO/IR; LiDAR is emerging.
- **Hard problems:** Tiny RCS, low thermal-signature airframes, urban clutter and multipath remain tough; the survey's challenge section calls out IR-low targets and argues for fused visible/IR + acoustic as mitigation.

Metric	Target (typical)
Probability of Detection (Pd)	> 0.95 for Class I/II UAS within designated bubble
False Alarm Rate	< 1 per operating hour (after fusion)
Time-to-Detect	/ < 5 s / < 15 s in inner zone
Time-to-Engage	
Coverage & Availability	360° coverage, > 99.5% system availability
Deconfliction	ATC/ADS-B/Remote ID integration, geofencing
Safety	No unintended RF/laser hazards; certified ROE

- Multi-layered sensor fusion, AI-aided ID, scalable effectors
- Counter-swarm & low-signature detection remain open research